

F. Multimodal Inference and generation

F.1 Introduction to Multimodal Representation & Alignment

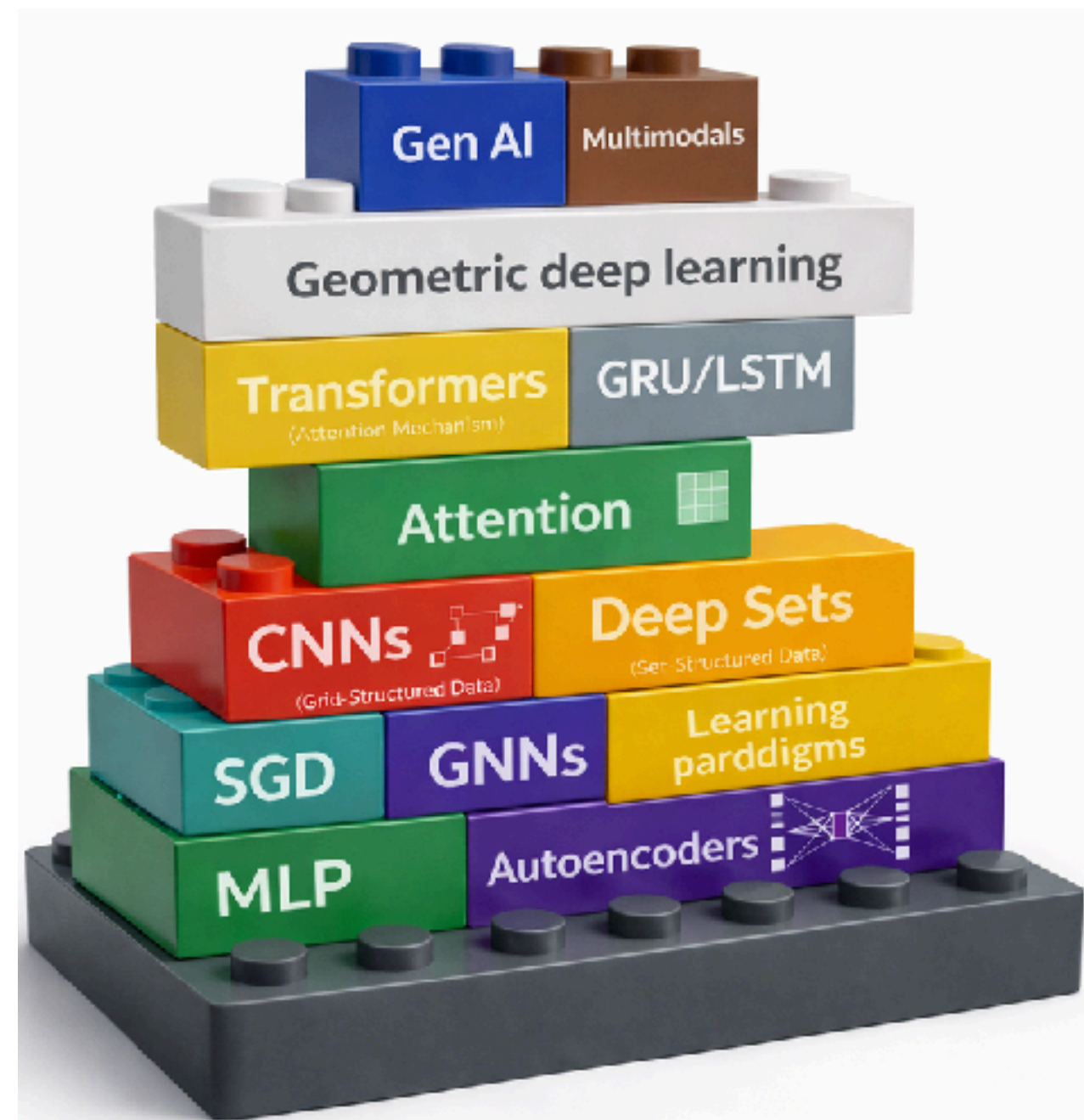
F.2 Multimodal Fusion

F.3 Cross-Modal learning



Tiam Heydari, PhD
Postdoctoral Fellow, UBC

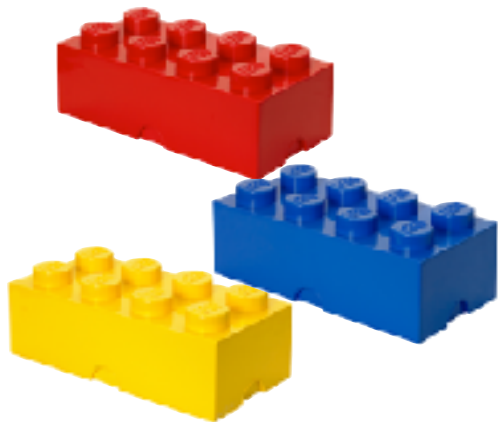
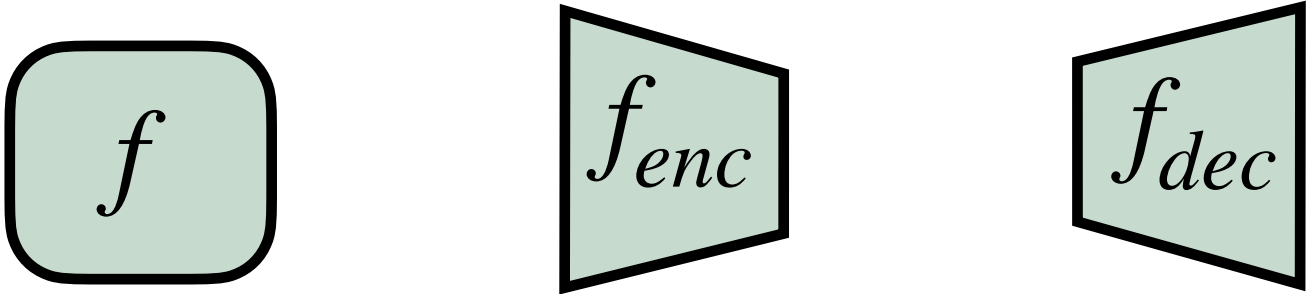
Before this last lecture, let's review what we learned in the course so far and try to put it in a coherent (ish) map



Input and output

$$X = \{ \triangle, \triangle, \dots \}$$
$$\hat{Y} = \{ \square, \square, \dots \}$$

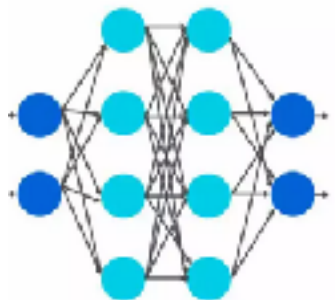
Neural network blocks:



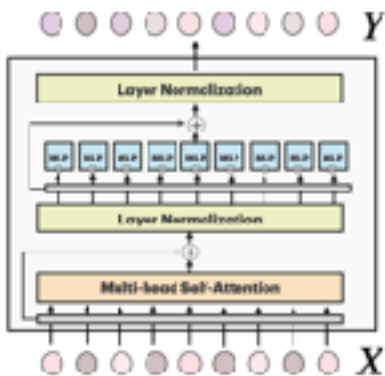
Fixed feature vector

$$x_i \in \mathbb{R}^d$$

MLP



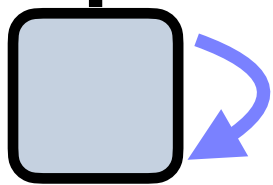
Transformer



Ordered sequence

$$x_i = (x_i^{(0)}, x_i^{(1)}, \dots, x_i^{(T_i-1)})$$

RNN



Transformer

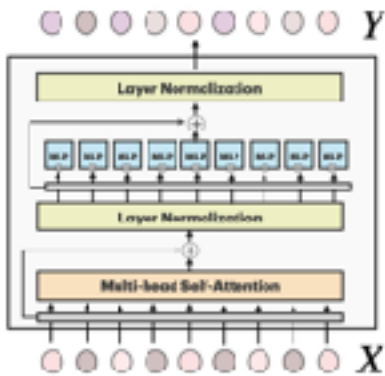
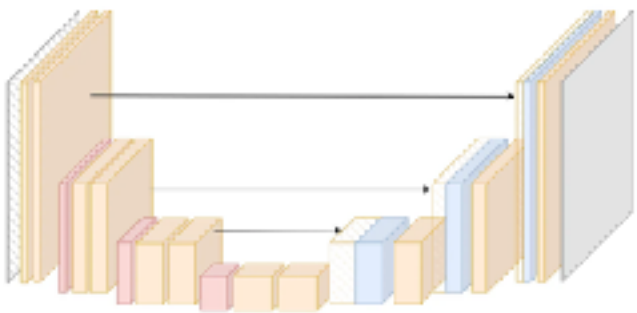


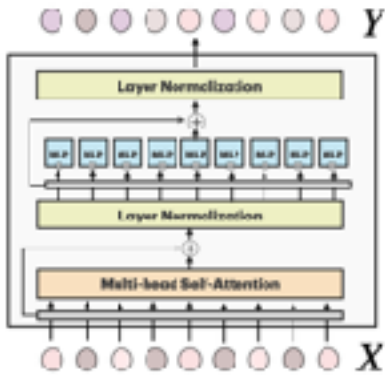
Image / spatial grid

$$x_i \in \mathbb{R}^{H_i \times W_i \times C}$$

CNN



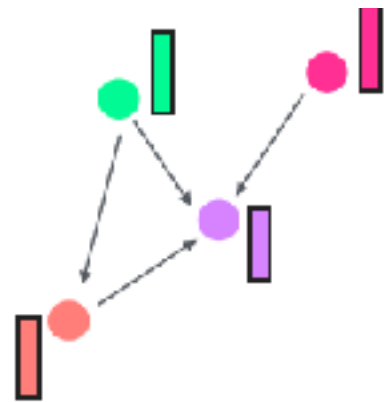
Transformer



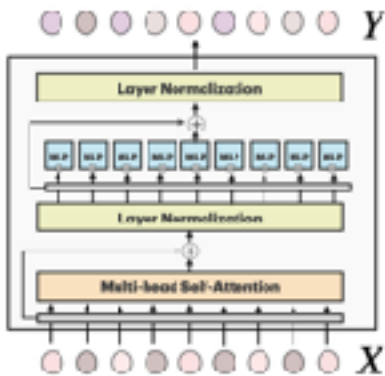
Graph

$$x_i = (V_i, E_i, \{x_{iv}\}_{v \in V_i})$$

GNN



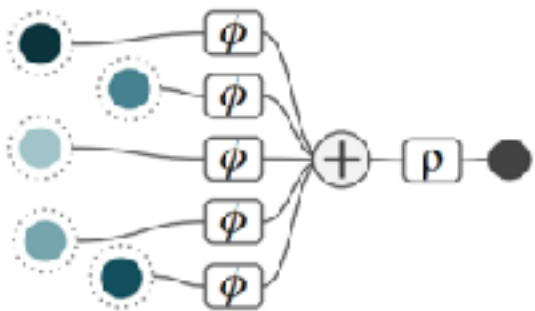
Transformer



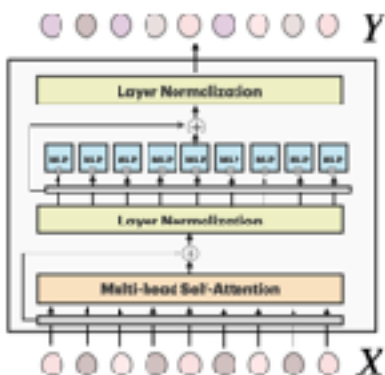
Unordered set

$$x_i = \{x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n_i)}\}$$

Deep Sets

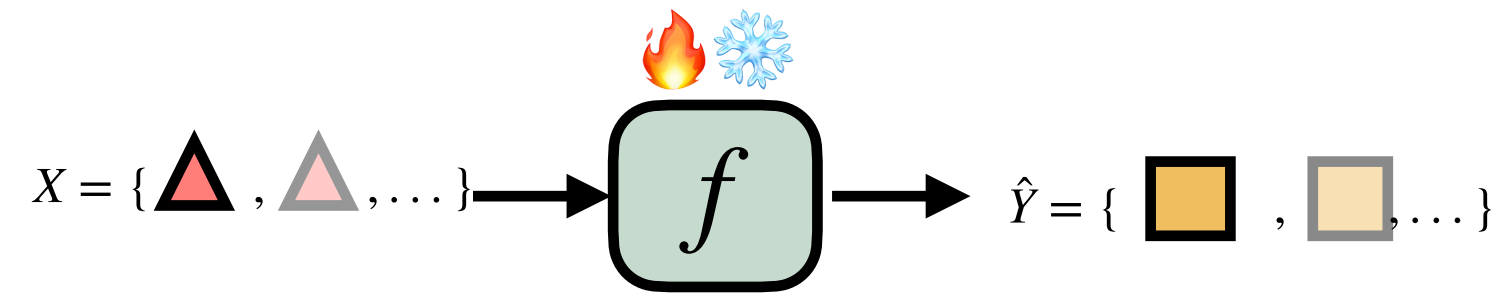


Transformer

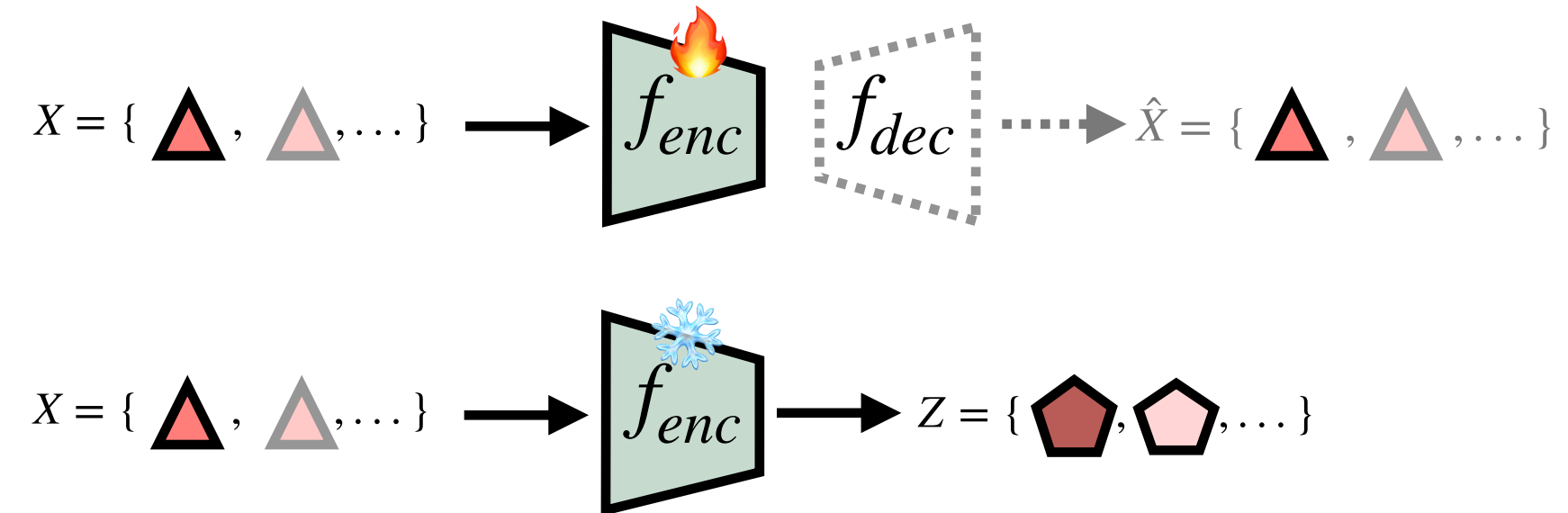




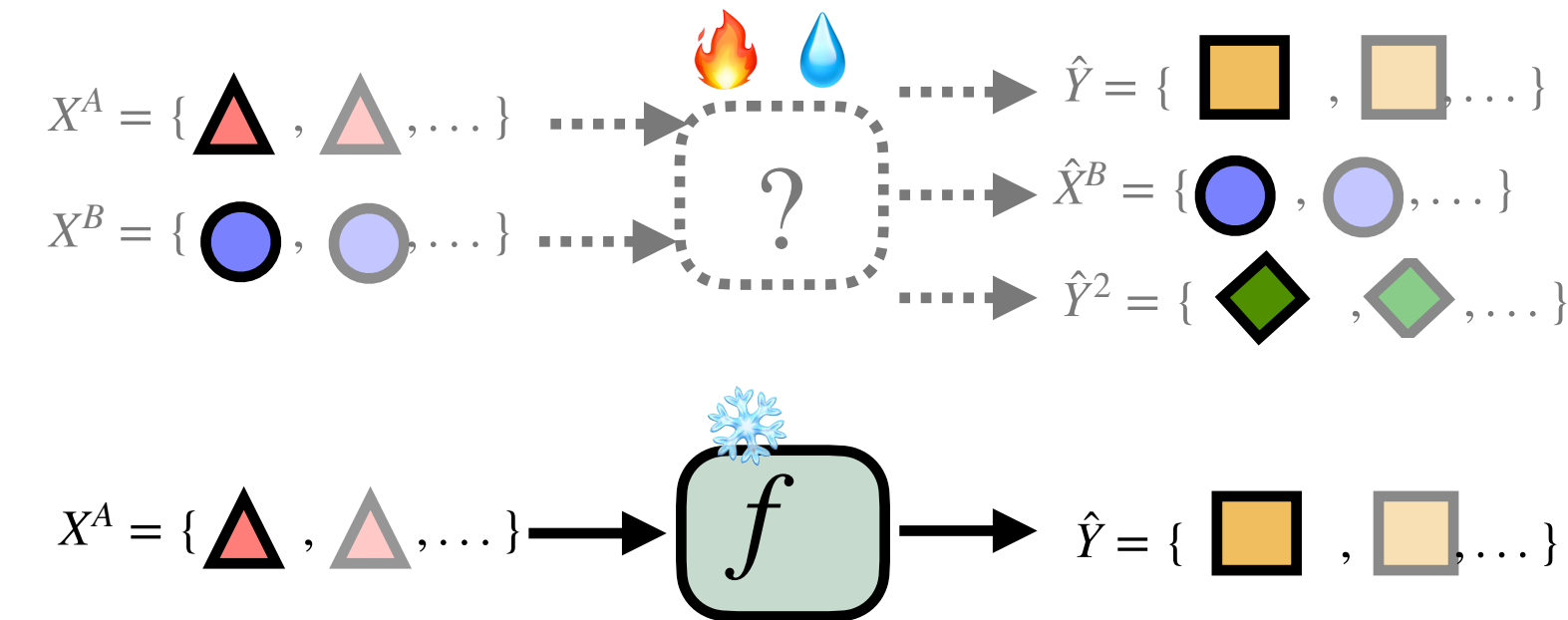
Supervised Learning



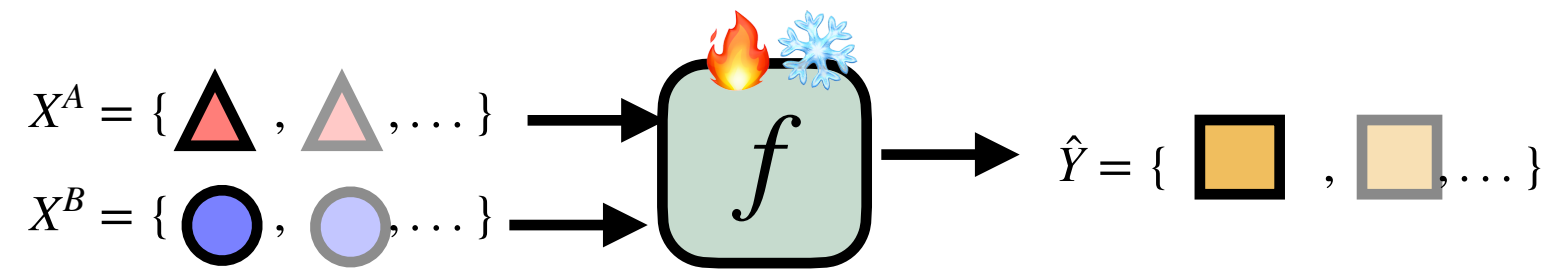
Self-Supervised Representation Learning



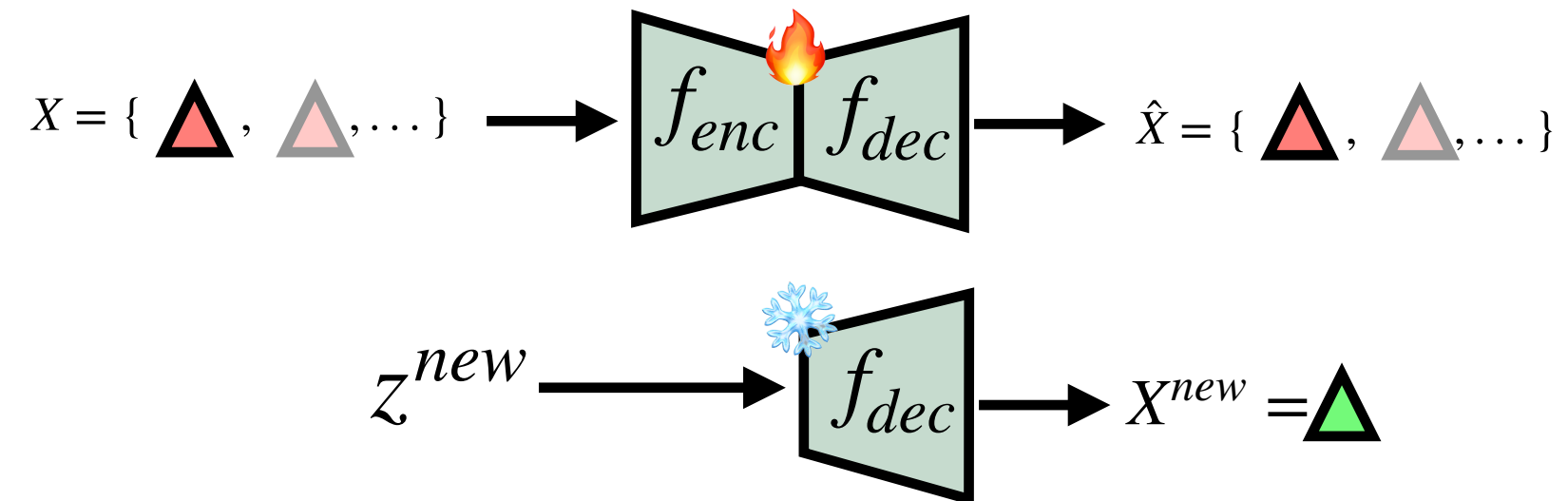
Transference (This lecture)



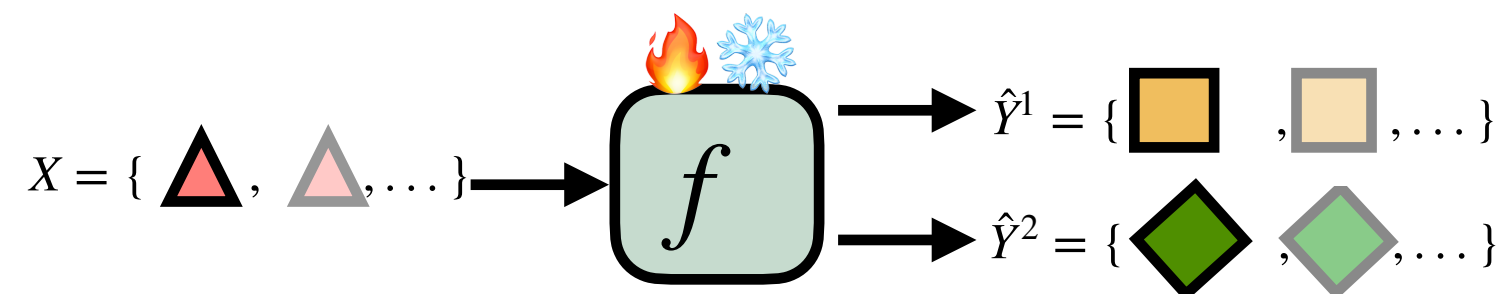
Multimodal Supervised Learning



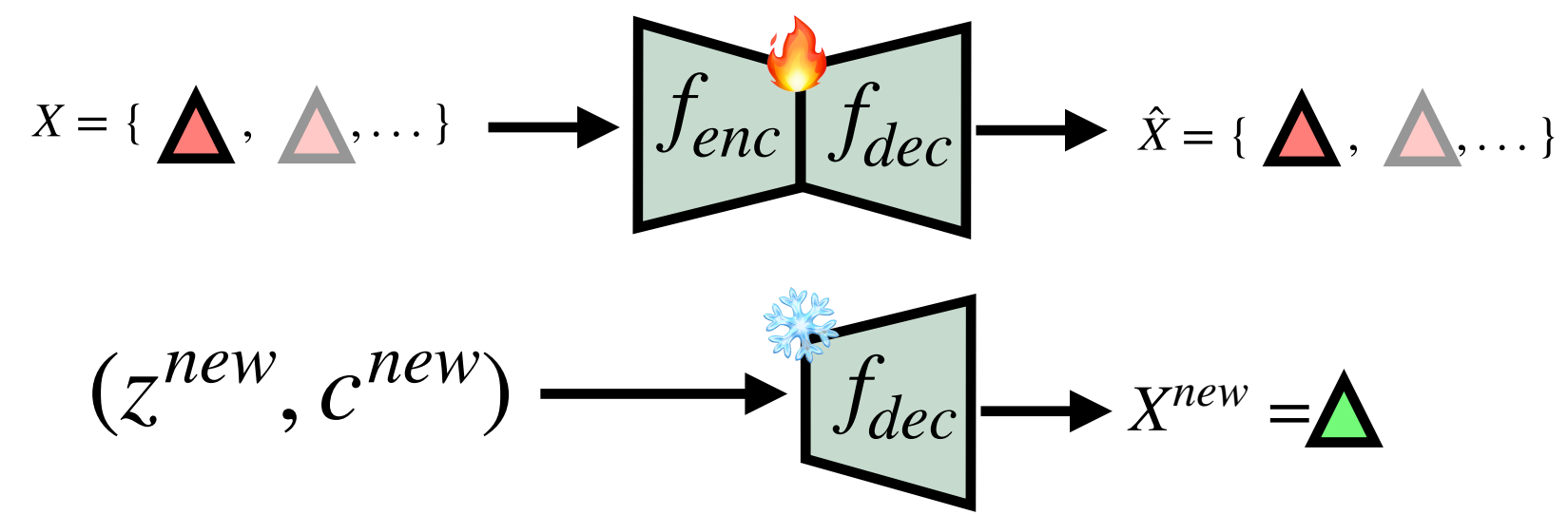
Unconditional Generative AI



Multi-task Supervised Learning



Conditional Generative AI



Training

Fine tuning

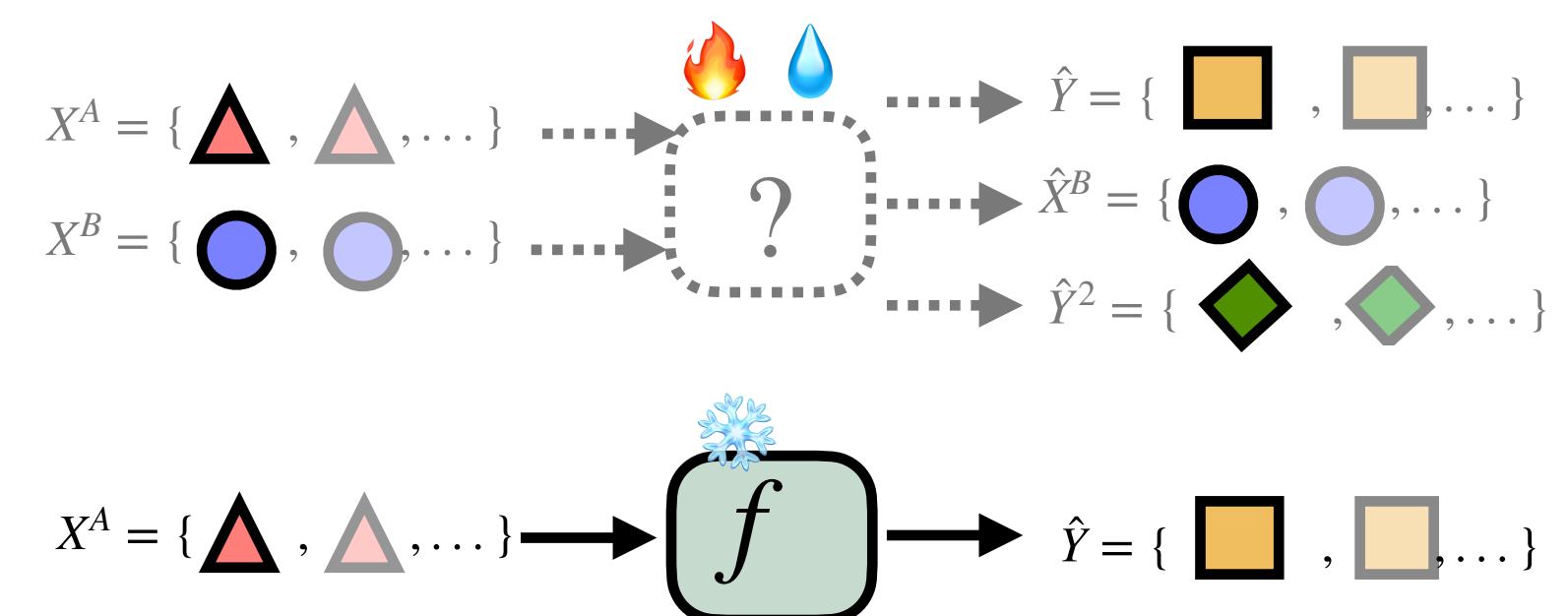
Deployment

Now, Transference



What is Transference ?

Transference (This lecture)

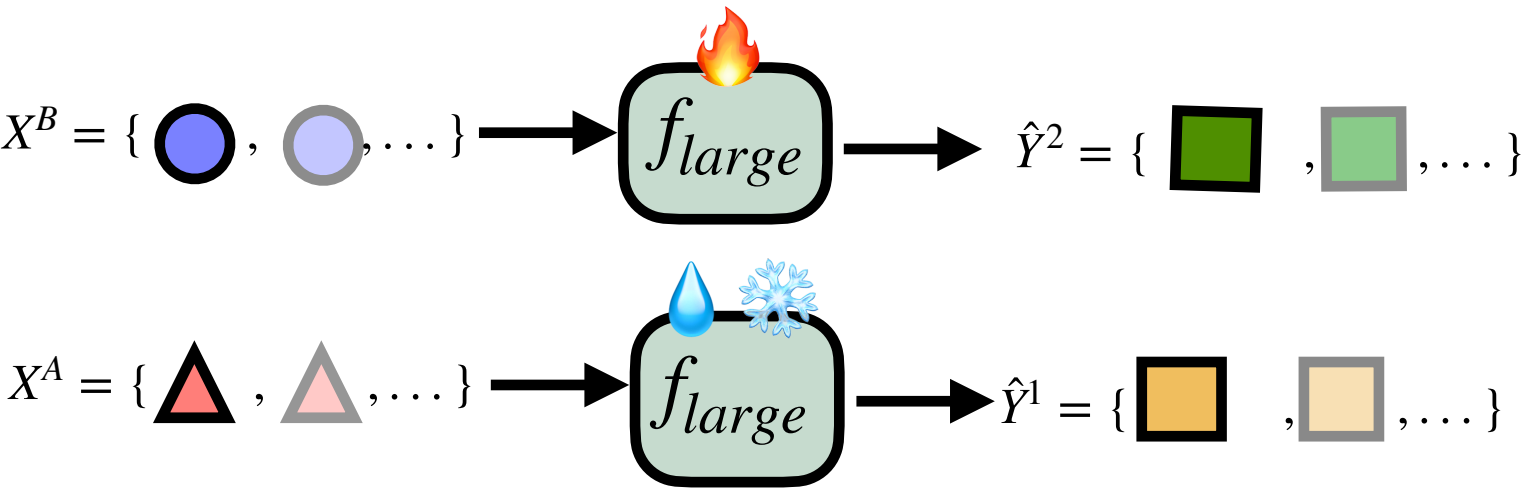


Transfer knowledge between modalities, or data groups, usually to help the Task in hand which may have noisy or with limited resources

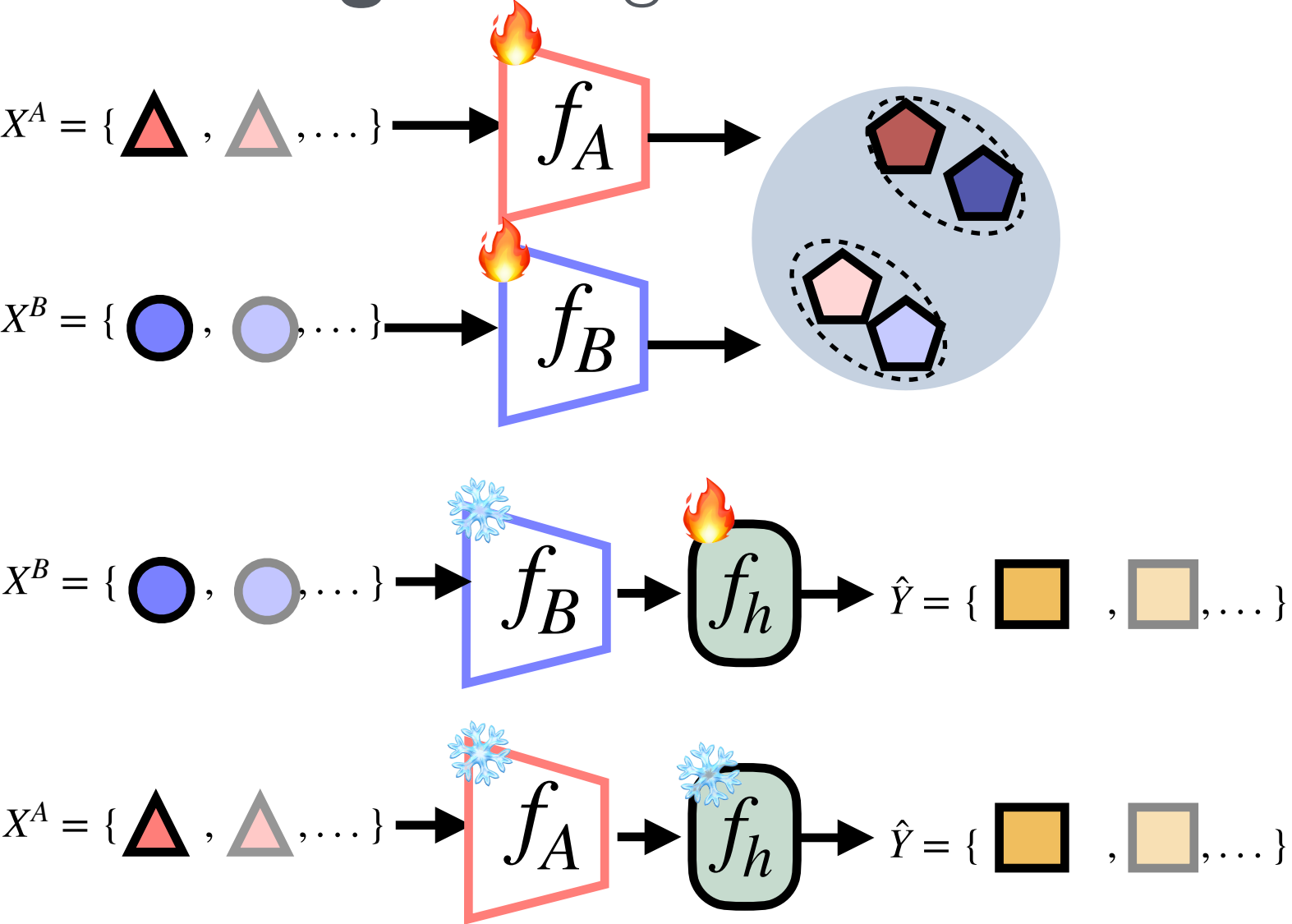


Multiple different combination and approaches go under the umbrella of Transference:

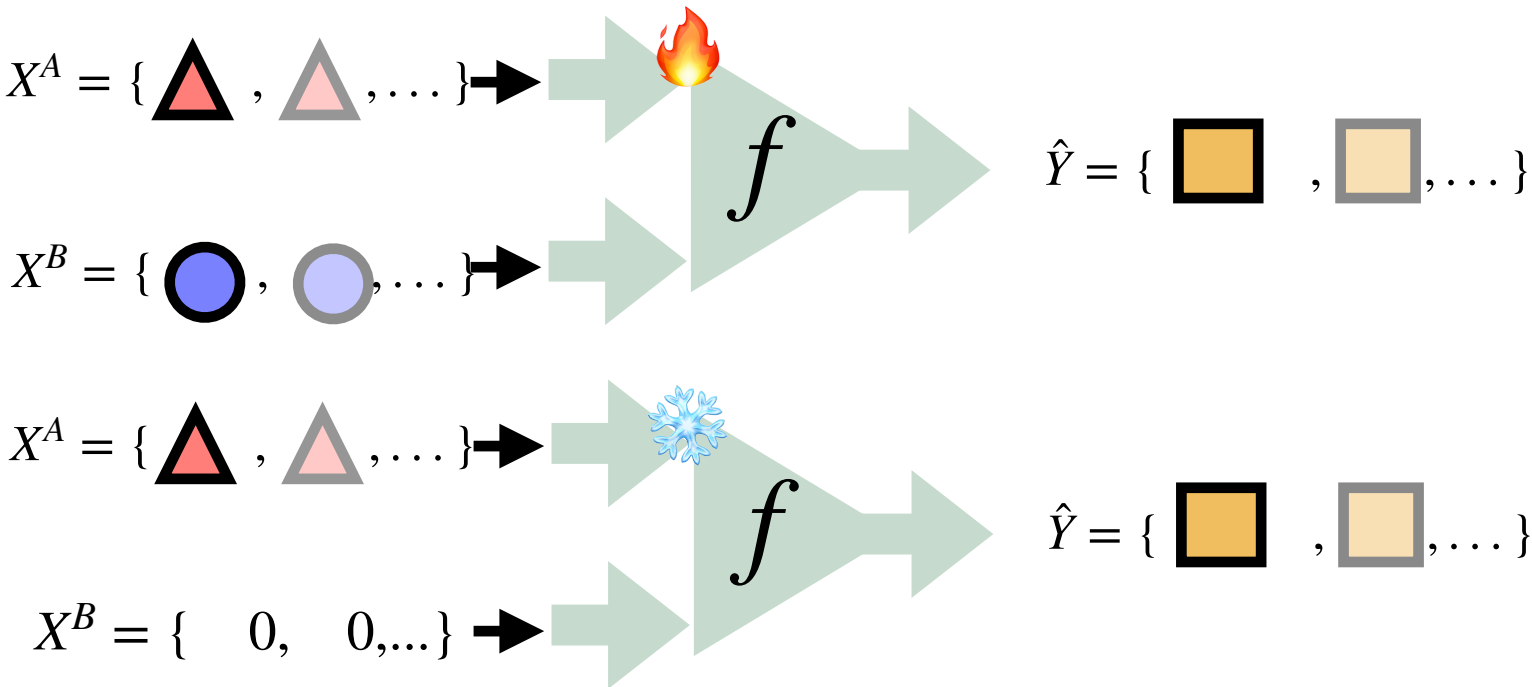
Transfer learning



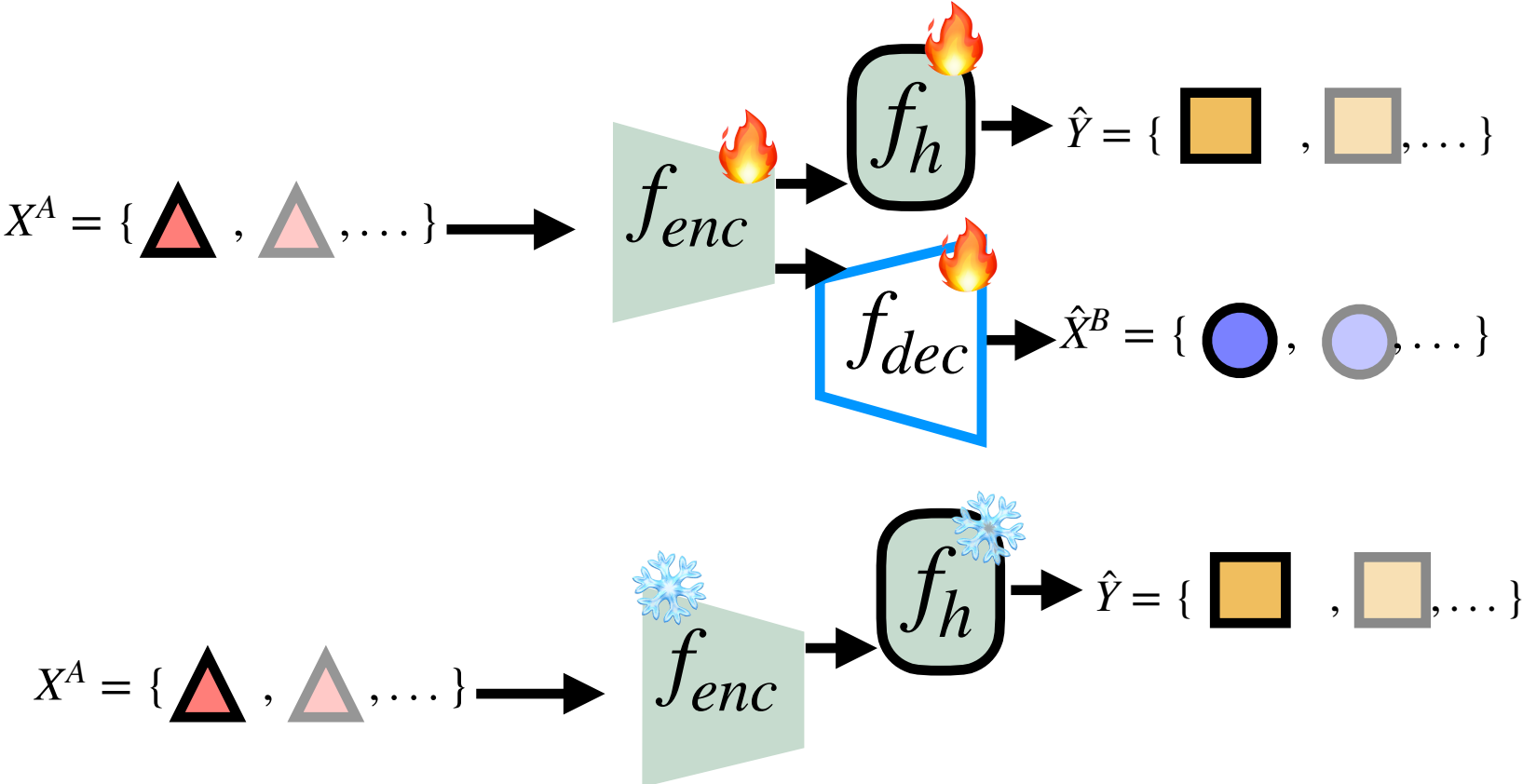
Co-learning via Alignment



Co-learning via Fusion



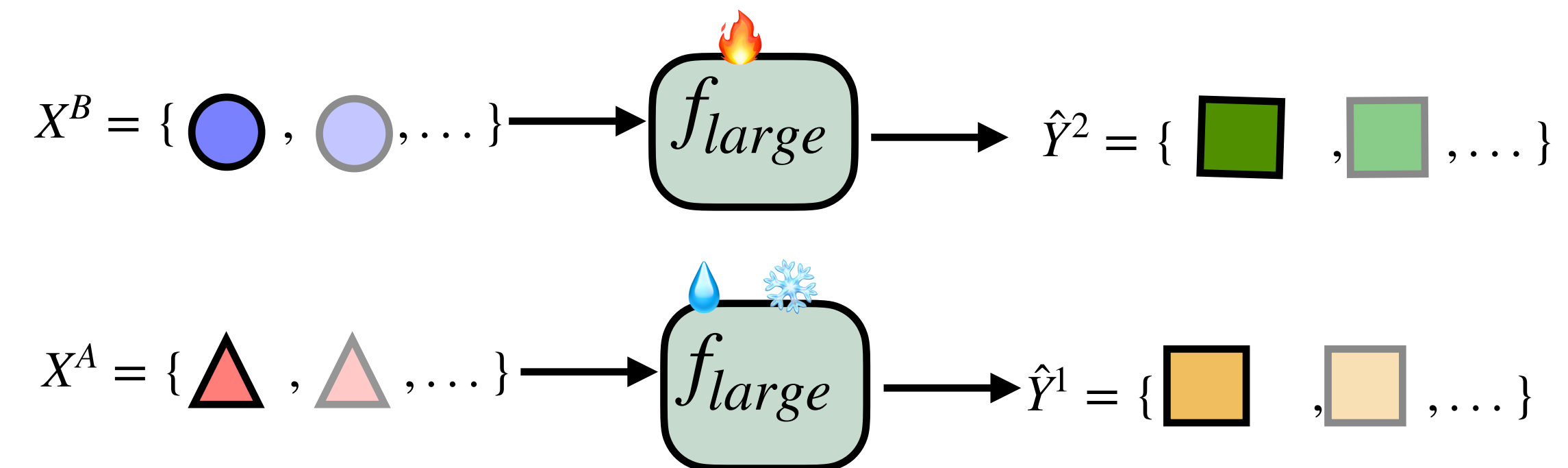
Co-learning via Translation



 Training
 Fine tuning
 Deployment



Transference: Transfer learning



Pretrain on a large, data-rich source task ; then fine-tune the learned model for a related target task with limited data. (*Note that training could be self supervised*)



Transference: Transfer learning

Example: Geneformer

 **Training**

Self-supervised large-scale pretraining

Genecorpus-30M



Self-supervised pretraining



 **Fine tuning**

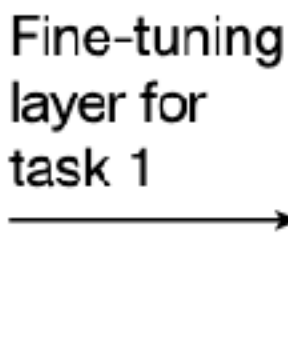
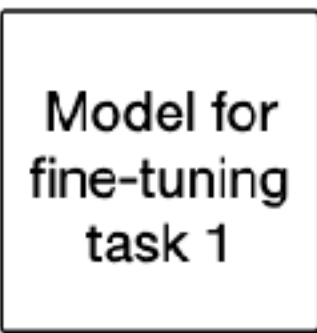
 **Deployment**

Fine-tuning with limited task-specific data



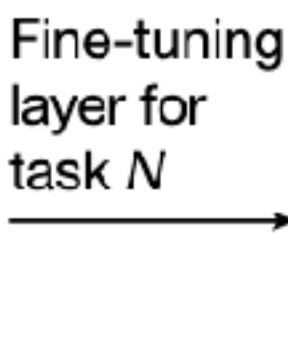
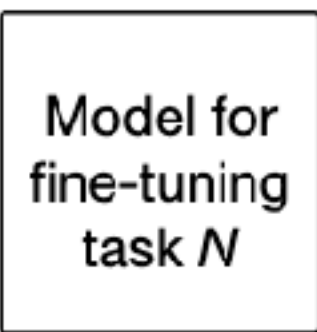
Limited task-specific data for task 1

Fine-tuning



Task 1 predictions

...



Task N predictions

Fine-tuning



Limited task-specific data for task N

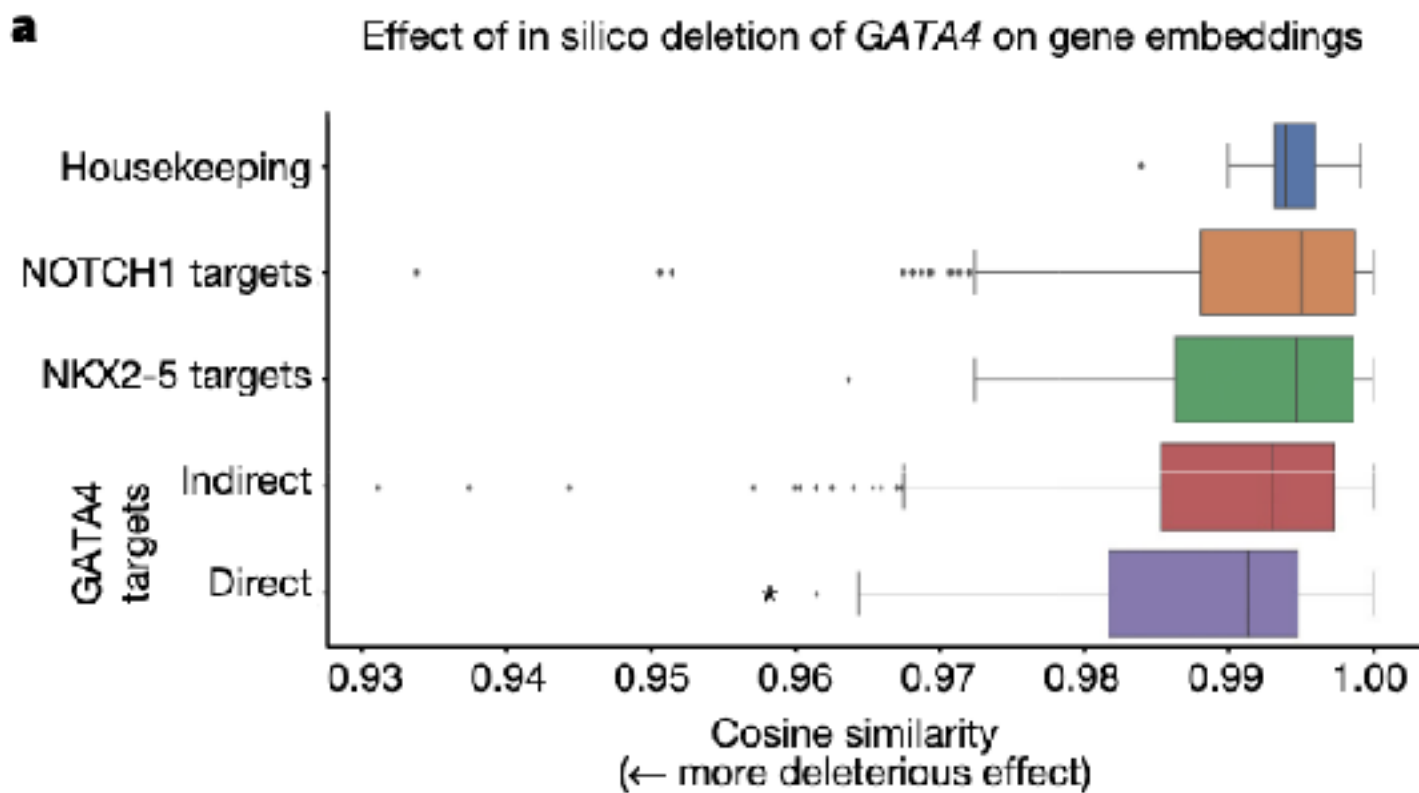
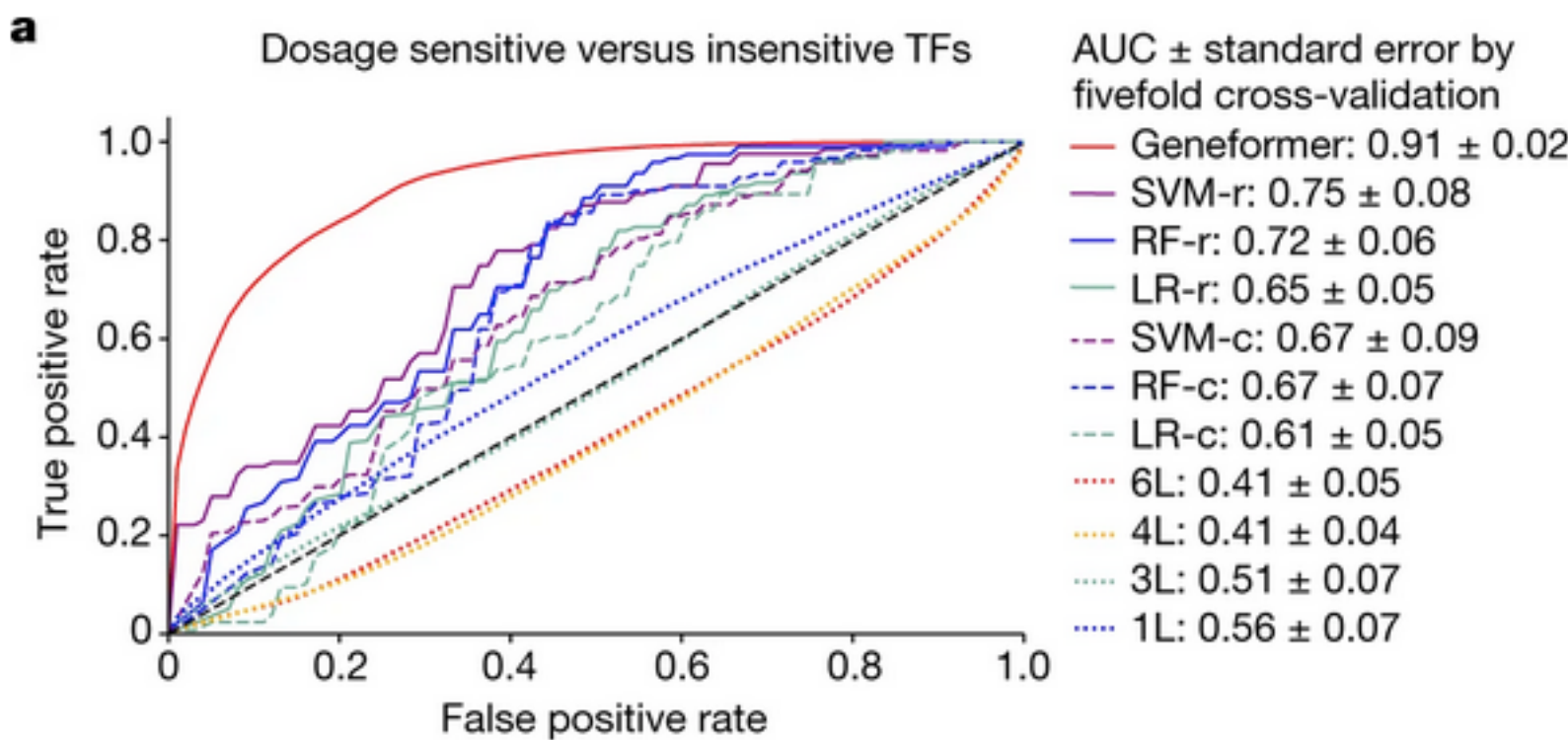


Fig. 5 | In silico deletion revealed network connections. **a**, In silico deletion



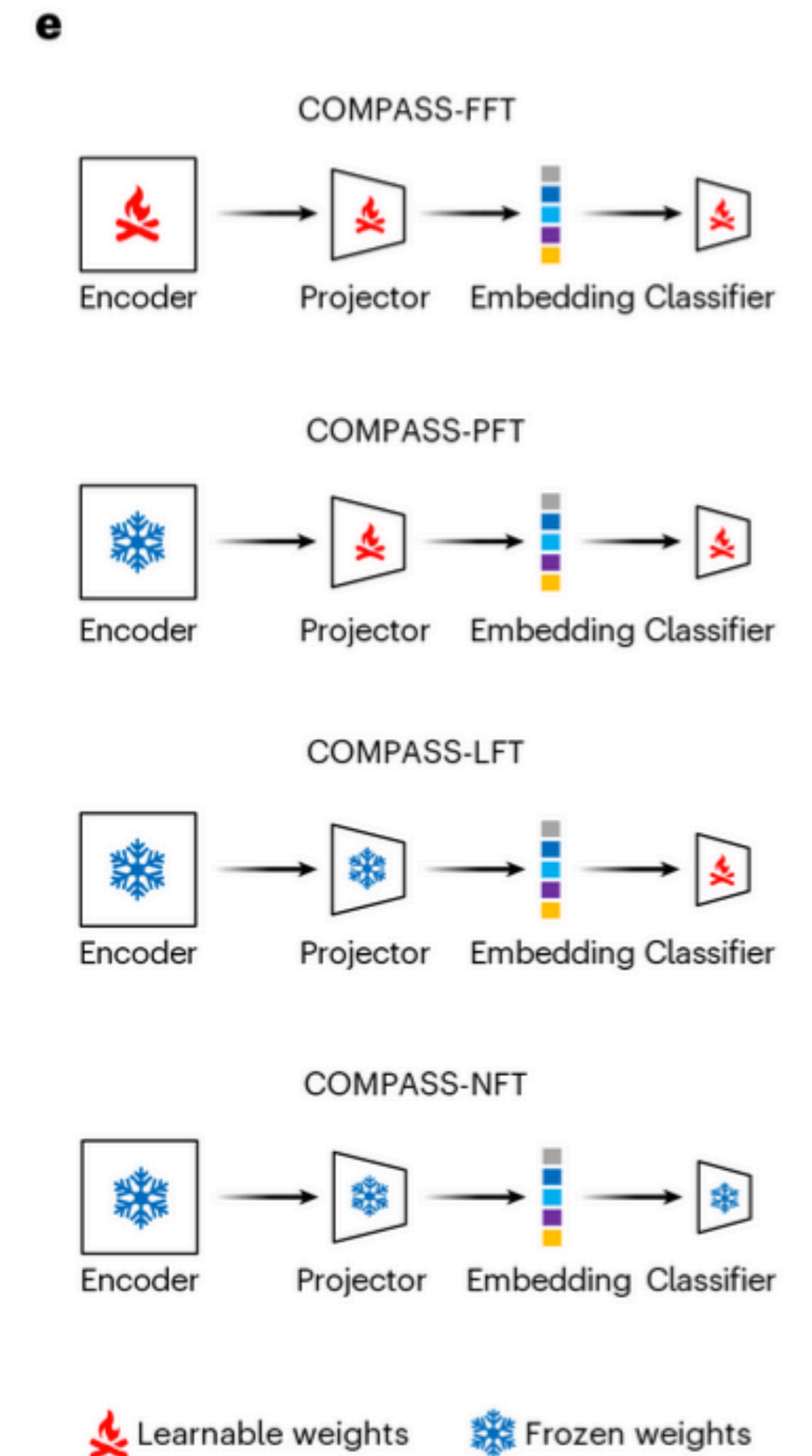
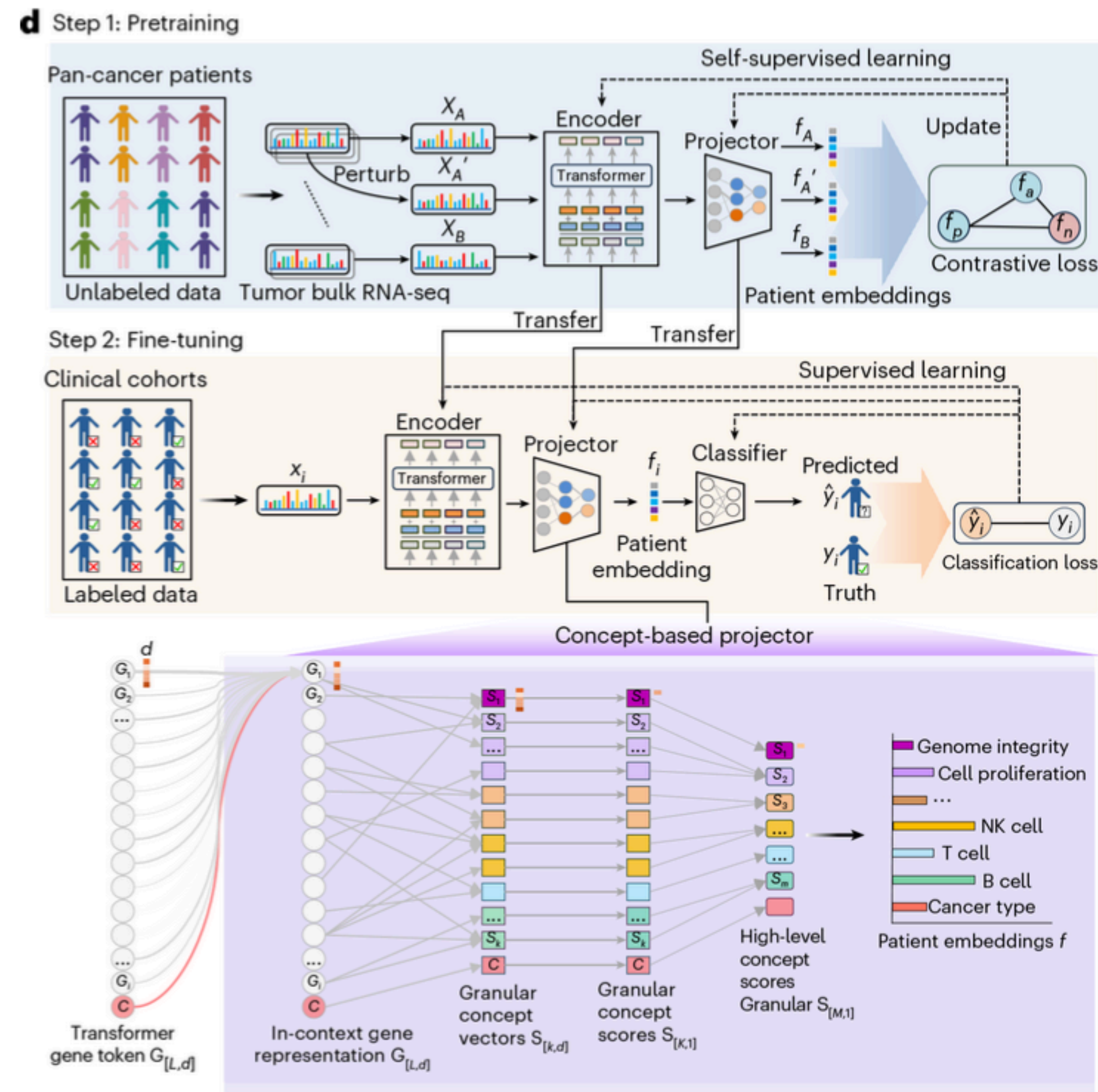
Example of combinations

COMPASS, multiple methods stacked in one pipeline: (Encoder is a transformer)

Self-supervised contrastive pretraining : learn a representation without labels.

Transfer / fine-tuning for clinical tasks : the three fire/frozen modes in panel (e)

Concept-bottleneck architecture for explainability : map to concepts before the classifier





Co-Learning: Use an **auxiliary modality** available during training to improve a model that will later operate from the primary modality alone.
It is helpful where the **auxiliary modality** is available only at training, and not at deployment.

primary

$$X^A = \{ \text{🔥} \triangle, \triangle, \dots \}$$

Primary

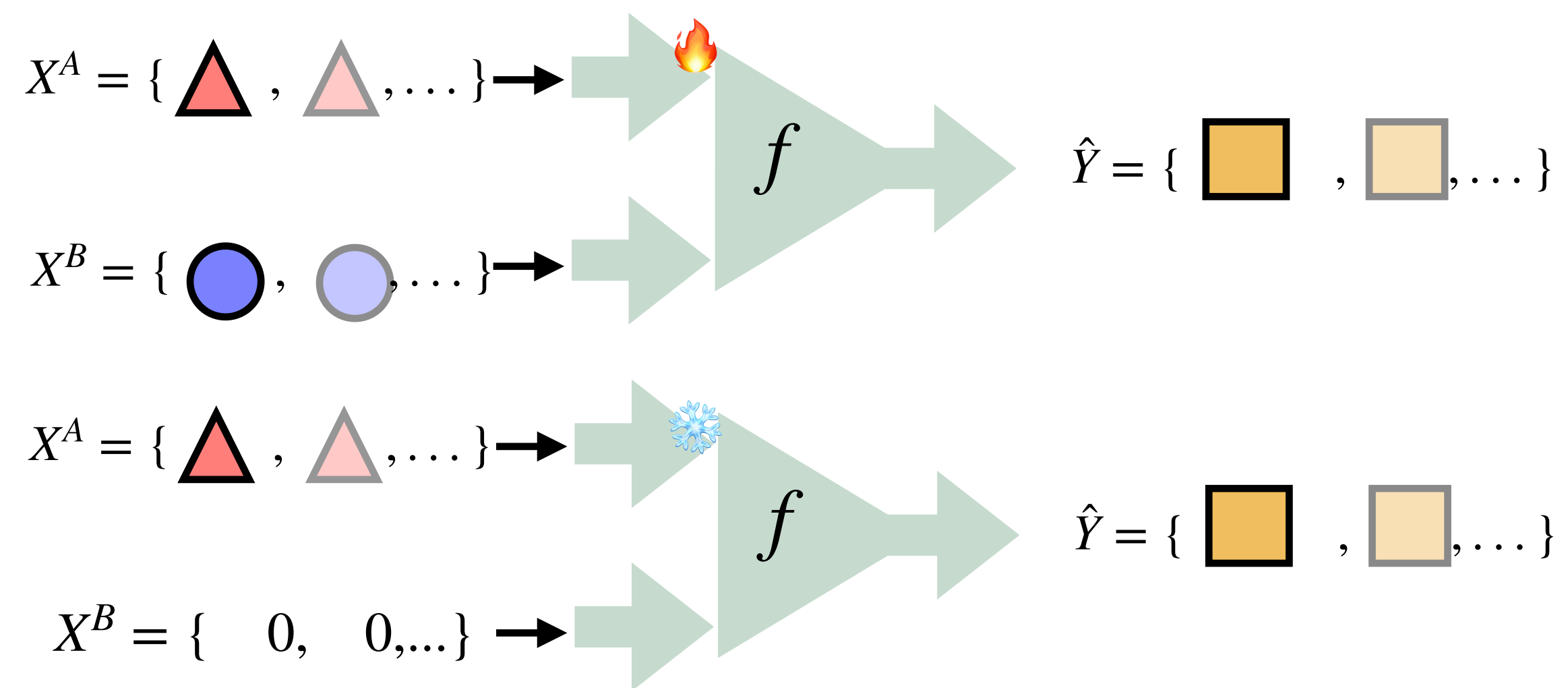
$$X^A = \{ \text{❄️} \triangle, \triangle, \dots \}$$

auxiliary

$$X^B = \{ \text{🟦}, \text{🟩}, \dots \}$$



Co-learning via Fusion



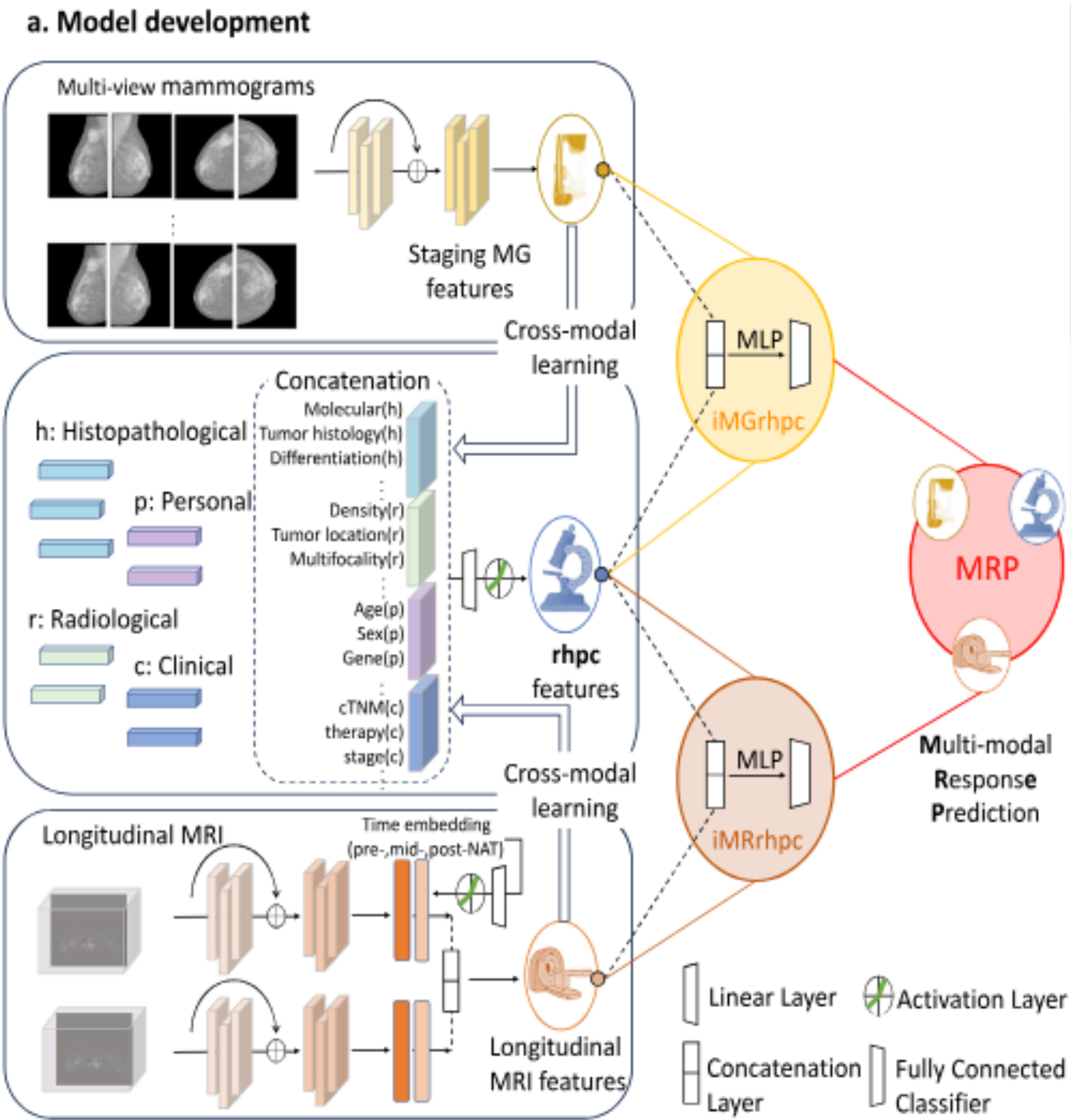
Fuse the primary (x^A) and auxiliary (x^B) modalities during training, but randomly hide the auxiliary modality for some training examples so the predictor learns to only use x^A when x^B is not available .

*Fusion is the bluntest way to do co-learning. Is usually mixed with other methods



Co-learning via Fusion

Example:



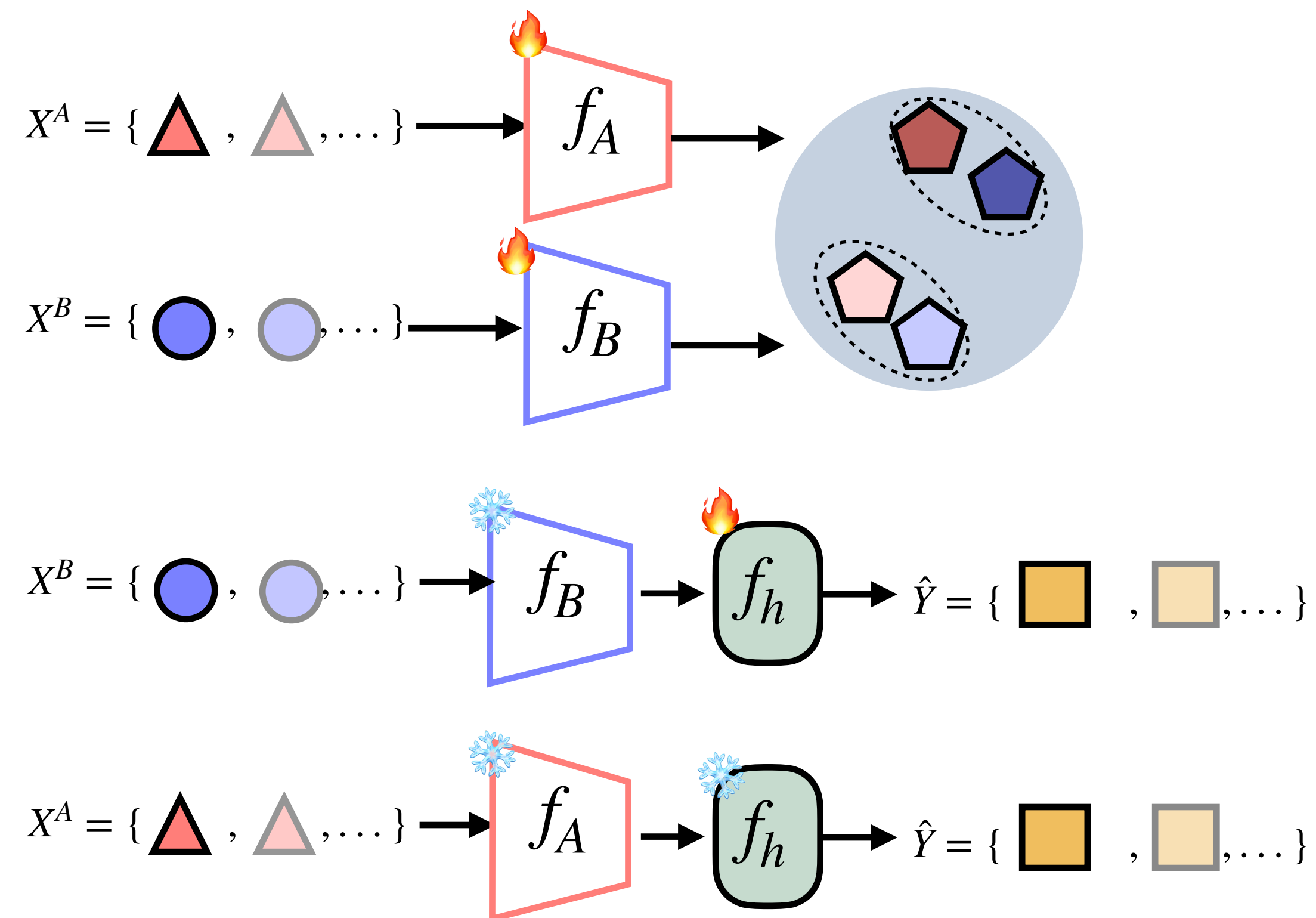
Model	In-house test set
rhpc	Pre-NAT
AUROC	0.75 (0.65-0.85)
AUPRC	0.60 (0.44-0.78)
Sensitivity	0.69 (0.39-0.89)
Specificity	0.73 (0.50-0.96)
iMGrhpc	Pre-NAT
AUROC	0.83 (0.76-0.91)
AUPRC	0.64 (0.49-0.80)
Sensitivity	0.82 (0.62-0.95)
Specificity	0.74 (0.59-0.90)
iMRrhpc	Pre-NAT
AUROC	0.84 (0.76-0.91)
AUPRC	0.67 (0.52-0.84)
Sensitivity	0.78 (0.60-0.92)
Specificity	0.78 (0.61-0.93)
PPV	0.64 (0.48-0.82)
MRP	Pre-NAT
AUROC	0.88 (0.82-0.94)
AUPRC	0.77 (0.63-0.90)
Sensitivity	0.82 (0.69-0.92)
Specificity	0.84 (0.74-0.93)
PPV	0.71 (0.57-0.85)

More modalities





Co-learning via Alignment



Learn paired encoders f_A and f_B that map corresponding observations from both modalities into one shared representation space. Then train a prediction head f_h in that space, so either modality alone can be used to predict $\hat{Y}$ at deployment.



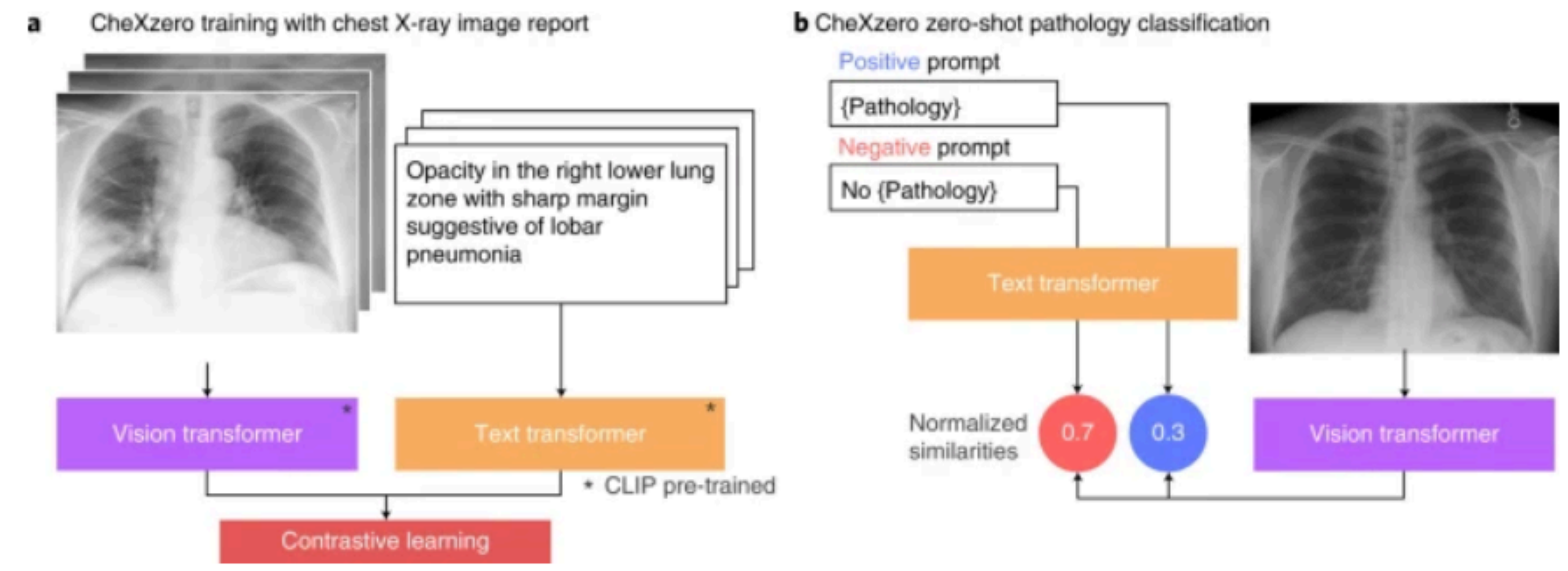
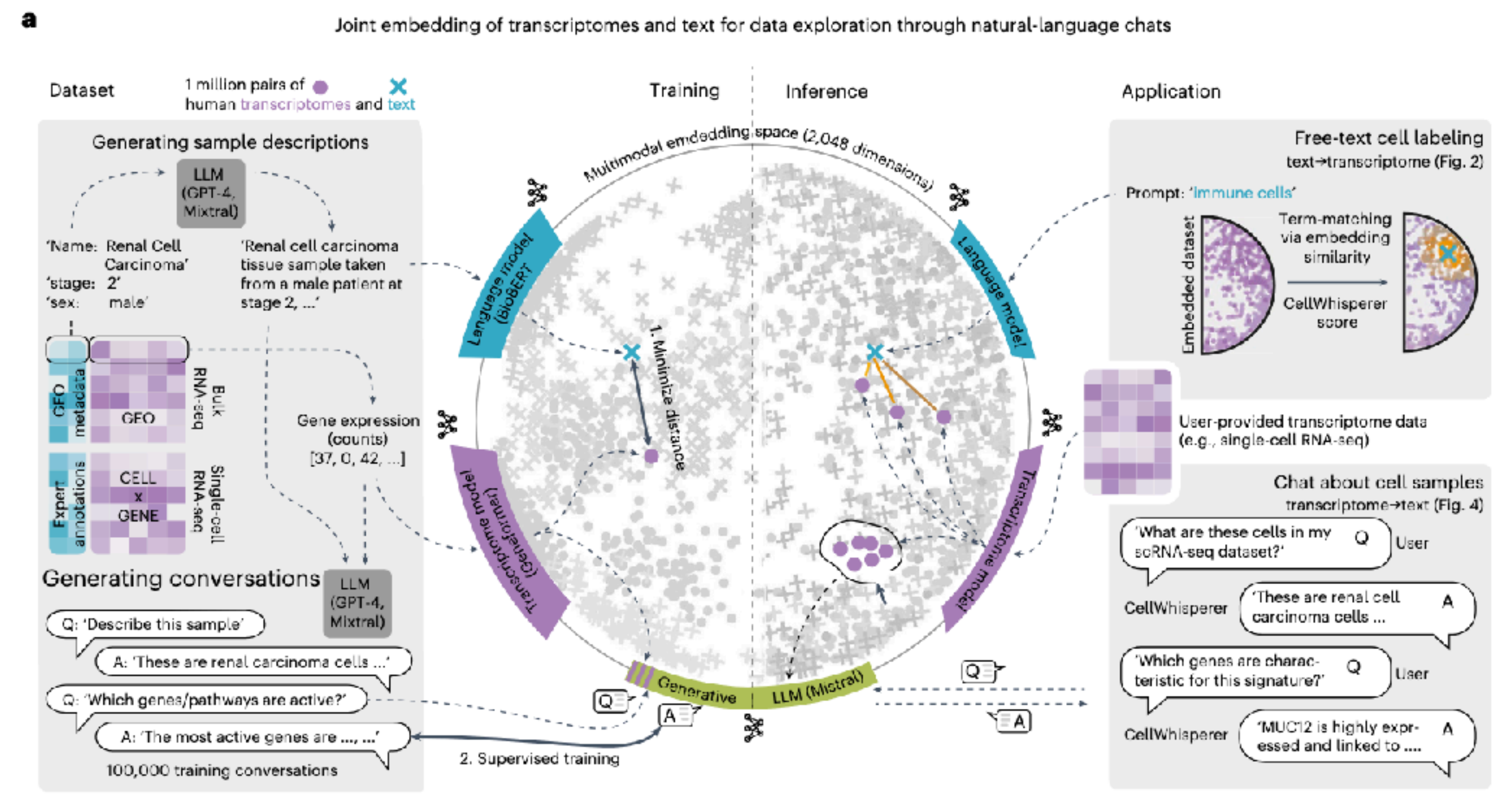
Co-learning via Alignment

Example: CellWhisperer: maps two modality-specific encoders to 2,048-dim space via alignment

Schaefer, Moritz, et al. "Multimodal learning enables chat-based exploration of single-cell data." *Nature Biotechnology* (2025):

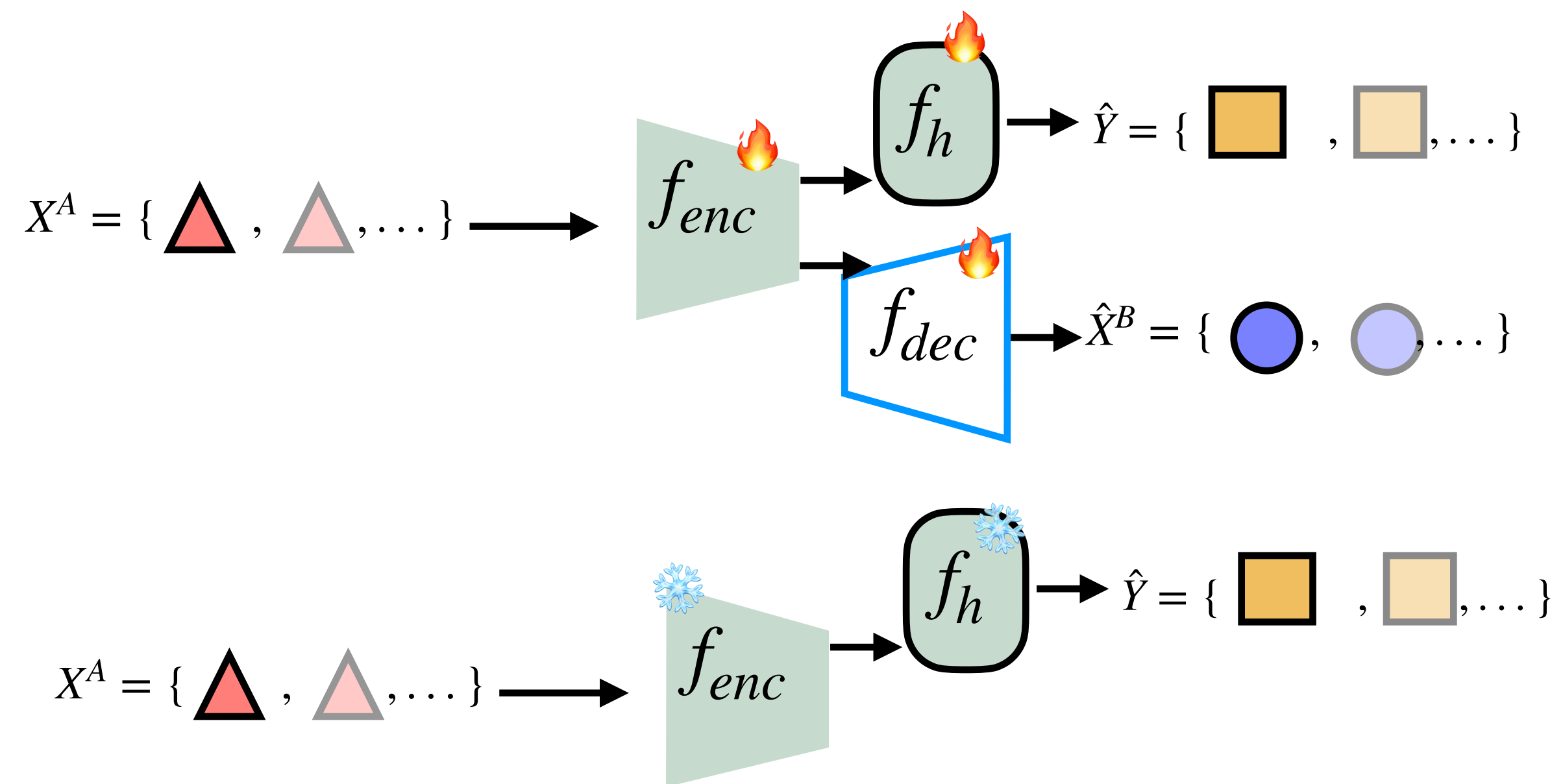
Example: An image encoder and a Transformer text encoder, first were jointly aligned on 377,110 chest-X-ray/report pairs.

Tiu, Ekin, et al. "Expert-level detection of pathologies from unannotated chest X-ray images via self-supervised learning." *Nature biomedical engineering* (2022):





Co-learning via Translation



Train an encoder to extract features from the primary modality that are useful for both the task and predicting the auxiliary modality. so the primary-modality representation must retain information about X^B .

At deployment, discard the decoder and predict from X_A alone.



Co-learning via Translation

Example:

🔥 Training

❄️ Deployment

$X^A = \{ \triangle, \triangle, \dots \}$

nocturnal breathing signal

$\hat{Y} = \{ \blacksquare, \blacksquare, \dots \}$

Disease detection:
has PD or not?

PD severity prediction
(Estimate of patient's
MDS-UPDRS)

$\hat{X}^B = \{ \bullet, \bullet, \dots \}$

qEEG

Extended Data Fig. 1: Nocturnal breathing data collection setup.

From: Artificial intelligence-enabled detection and assessment of Parkinson's disease using nocturnal breathing signals

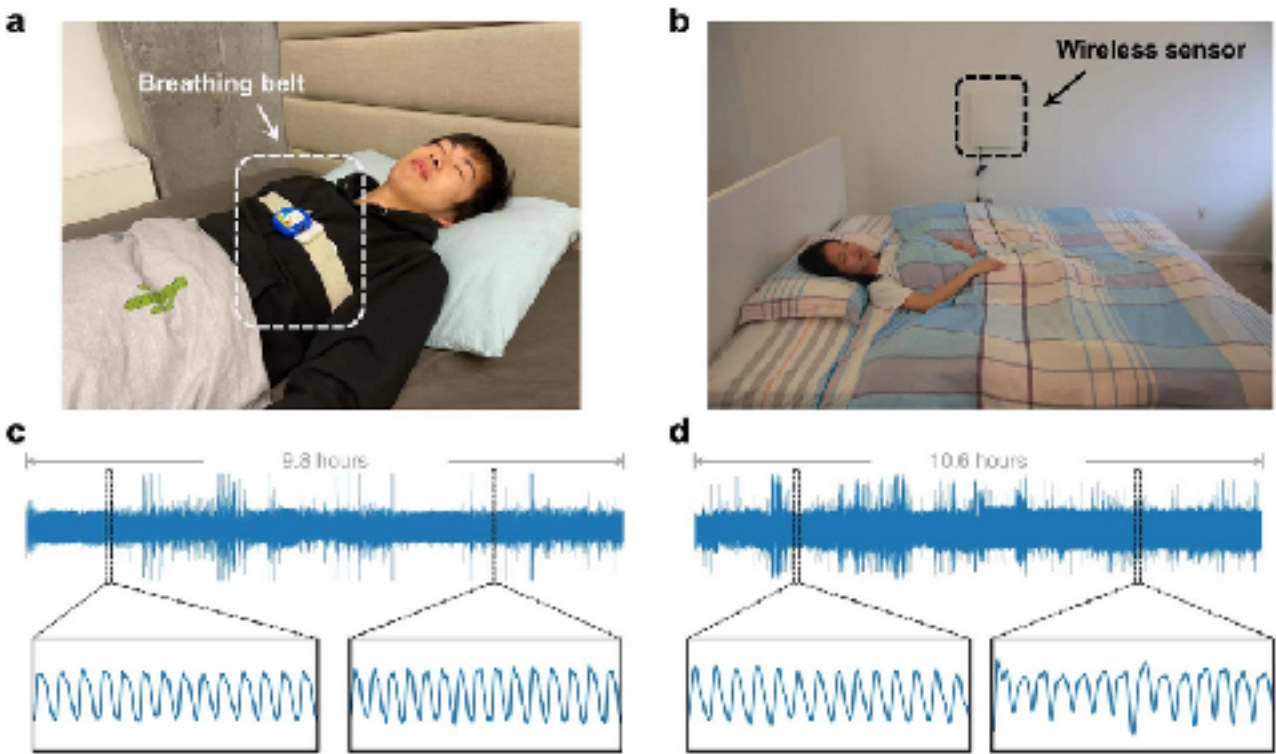
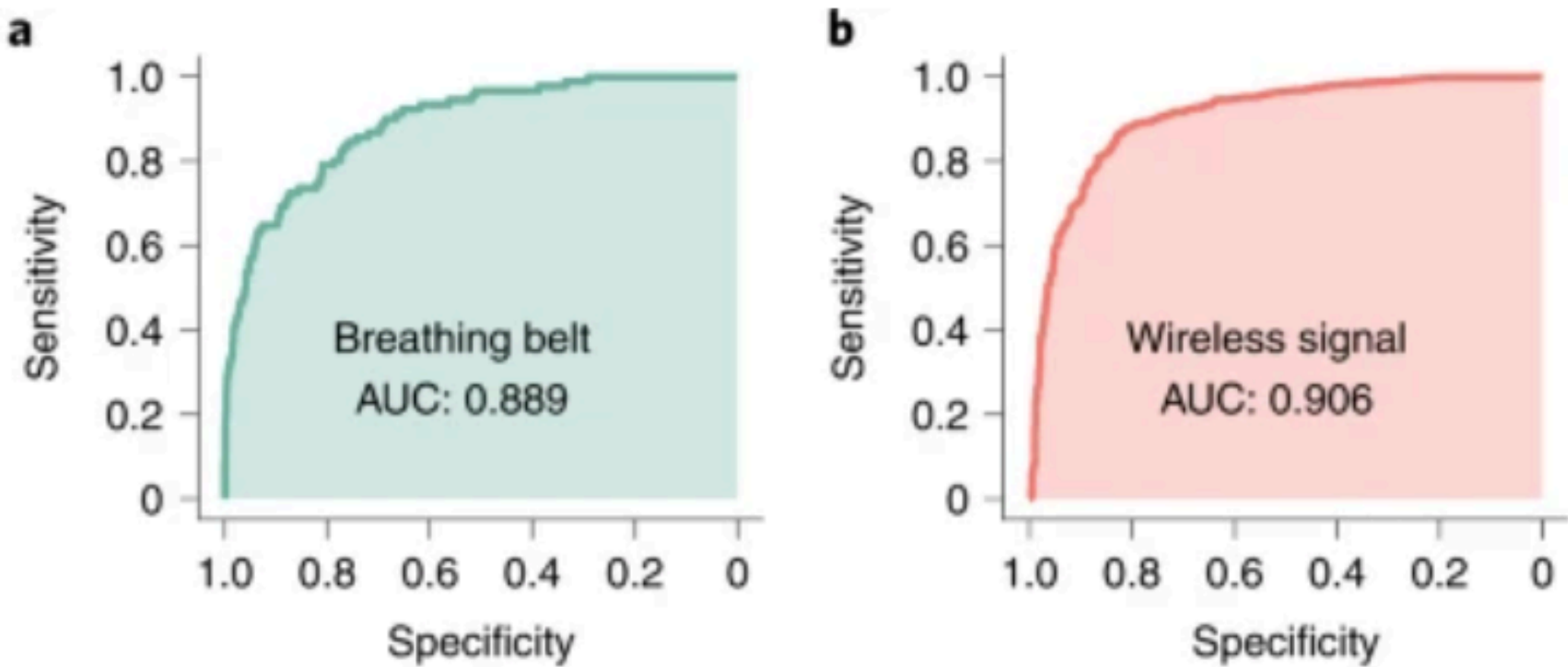
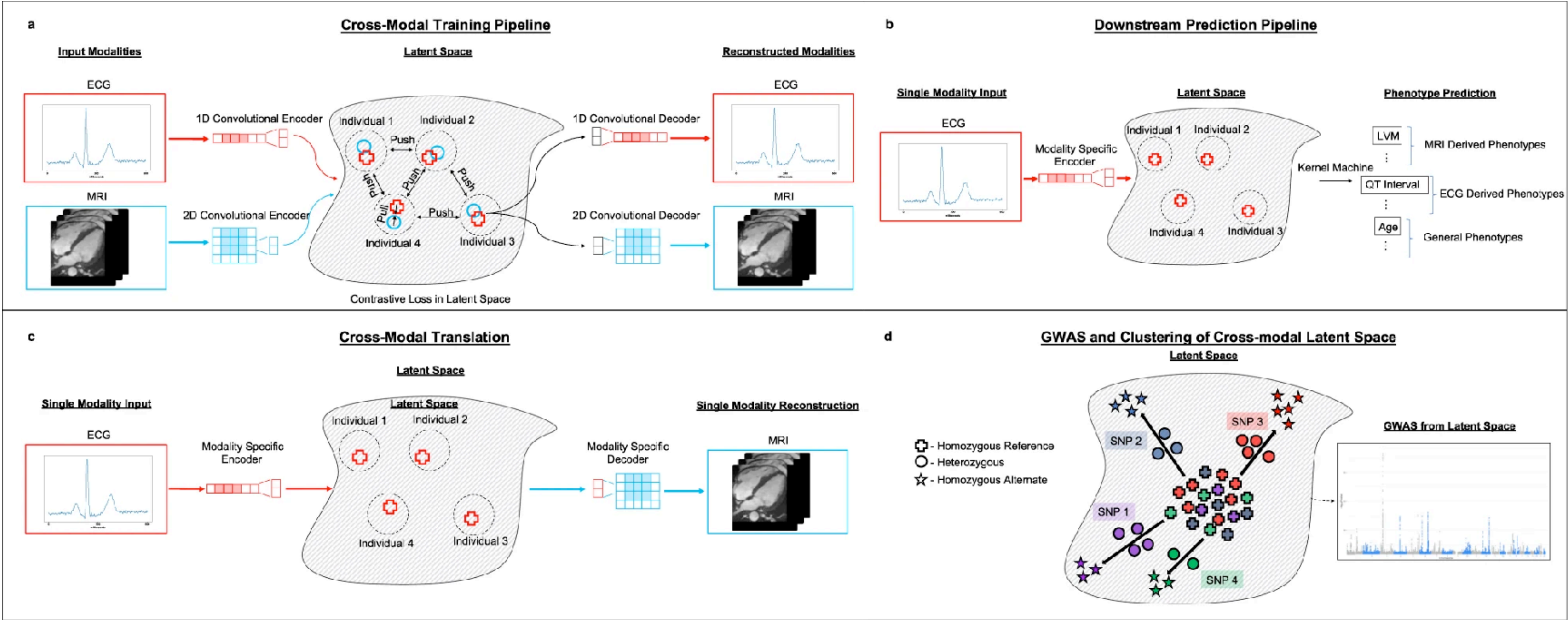


Fig. 2: PD diagnosis from nocturnal breathing signals.





Example of combinations





Read more:

Zhang, Sheng, et al. "Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs." *arXiv preprint arXiv:2303.00915* (2023).

Dong, Runpei, et al. "Dreamllm: Synergistic multimodal comprehension and creation." *International Conference on Learning Representation* Vol. 2024. 2024.

Turk, Matthew. "Multimodal interaction: A review." *Pattern recognition letters* 36 (2014): 189-195.

Liang, Paul Pu, et al. "Quantifying & modeling multimodal interactions: An information decomposition framework." *Advances in Neural Information Processing Systems* 36 (2023): 27351-27393.

Team, Chameleon. "Chameleon: Mixed-modal early-fusion foundation models." *arXiv preprint arXiv:2405.09818* (2024)

<https://cmu-mmml.github.io/fall2020/readings/>

<https://cmu-mmml.github.io/spring2023/>